

Analysis III - CheatSheet

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Regular Curve. A continuously differentiable map $\gamma : [a, b] \rightarrow \mathbb{R}^n$ is *regular* if $\gamma'(t) \neq 0$ for all $t \in [a, b]$.

Simple Curve. A continuous map $\gamma : [a, b] \rightarrow \mathbb{R}^n$ is *simple* if it does not intersect itself (except possibly at endpoints): $\gamma(t_1) = \gamma(t_2) \implies t_1 = t_2$

Simply Connected Domain. An open set $\Omega \subseteq \mathbb{R}^n$ is *simply connected* if for any two continuous curves $\gamma_0, \gamma_1 : [a, b] \rightarrow \Omega$ with $\gamma_0(a) = \gamma_1(a)$ and $\gamma_0(b) = \gamma_1(b)$, there exists a continuous homotopy $\Gamma : [a, b] \times [0, 1] \rightarrow \Omega$ that deforms γ_0 into γ_1 within Ω while keeping the endpoints fixed.

Curve in $\mathbb{R}^n$: Given $\gamma : [a, b] \rightarrow \mathbb{R}^n$, $\int_{\gamma} F \cdot d\mathbf{l} = \int_a^b \langle F(\gamma(t)), \gamma'(t) \rangle dt$, $\int_{\gamma} f(\mathbf{x}) ds = \int_a^b f(\gamma(t)) \|\gamma'(t)\| dt$.

$$\begin{aligned} \nabla \cdot (\vec{A} \times \vec{B}) &= (\nabla \times \vec{A}) \cdot \vec{B} - \vec{A} \cdot (\nabla \times \vec{B}) \\ \operatorname{div}(\nabla f) &= \Delta f \\ \operatorname{rot}(\nabla f) &= 0 \\ \operatorname{div}(\operatorname{rot} \vec{F}) &= 0 \quad \nabla(fg) = f \nabla g + g \nabla f \\ \operatorname{rot}(f \vec{F}) &= \nabla f \times \vec{F} + f \operatorname{rot}(\vec{F}) \\ \operatorname{div}(f \vec{F}) &= f \operatorname{div}(\vec{F}) + \langle \vec{F}, \nabla f \rangle \end{aligned}$$

Surface in $\mathbb{R}^3$: Given $\sigma : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$, $\sigma(u, v)$,

$$\iint_{\sigma} F \cdot d\mathbf{S} = \iint_D \langle F(\sigma(u, v)), \frac{\partial \sigma}{\partial u} \times \frac{\partial \sigma}{\partial v} \rangle du dv. \quad \iint_{\sigma} f(\mathbf{x}) ds = \iint_D f(\sigma(u, v)) \left\| \frac{\partial \sigma}{\partial u} \times \frac{\partial \sigma}{\partial v} \right\| du dv.$$

Conservative Fields and Path Independence

A map $F : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ is *conservative* if there exists a differentiable function $\varphi : \Omega \rightarrow \mathbb{R}$ such that $\nabla f = F$.
 F is conservative on $\Omega \iff \forall \Gamma_1, \Gamma_2 \subseteq \Omega$ reg. curves from A to B , $\int_{\Gamma_1} F \cdot d\mathbf{l} = \int_{\Gamma_2} F \cdot d\mathbf{l}$, $\iff \forall \Gamma \subseteq \Omega$ reg. closed curve, $\int_{\Gamma} F \cdot d\mathbf{l} = 0$.

is F Conservative over Ω ?

1 - Compute curl F. **2 - is Ω Simply Connected ?**

- If $\operatorname{curl} \mathbf{F} \neq \mathbf{0}$, $\mathbf{F}$ is **not** conservative.
- If $\operatorname{curl} \mathbf{F} = \mathbf{0}$, proceed to Step 2.
- If *yes*, $\mathbf{F}$ is conservative.
- If *no*, proceed to Step 3.

$$\mathbf{2D:} \nabla \times \mathbf{F} = \begin{bmatrix} \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \end{bmatrix}$$

$$\mathbf{3D:} \nabla \times \mathbf{F} = \begin{bmatrix} \frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \\ \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \\ \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \end{bmatrix}$$

3 - Circulation Method:

1. Select closed curves Γ around each hole in Ω .
2. Compute $\oint_{\Gamma} \mathbf{F} \cdot d\mathbf{l}$.
3. If any integral $\neq 0$, $\mathbf{F}$ is **not** conservative.
4. If all integrals = 0, proceed to Step 4.

4 - Finding the Potential

1. Determine $\alpha(y, z)$ such that $\nabla \varphi = \mathbf{F}$.
2. If successful, φ is the potential function for $\mathbf{F}$, confirming $\mathbf{F}$ is conservative.

$$\gamma(t) = (x(t), y(t)), \mathbf{n}(t) = \frac{(y'(t), -x'(t))}{\|\gamma'(t)\|} \quad \sigma(u, v) = (x(u, v), y(u, v), z(u, v)), \mathbf{n}(u, v) = \frac{\frac{\partial \sigma}{\partial u} \times \frac{\partial \sigma}{\partial v}}{\left\| \frac{\partial \sigma}{\partial u} \times \frac{\partial \sigma}{\partial v} \right\|}, \text{ !not necessarily outward}$$

Vector Calculus Theorems

Green's Theorem (Plane) Let $\Omega \subseteq \mathbb{R}^2$ be a regular domain with a positively oriented boundary $\partial\Omega$, and $F \in C^1(\overline{\Omega}, \mathbb{R}^2)$. Then

$$\iint_{\Omega} \operatorname{curl}(F(x, y)) dx dy = \int_{\partial\Omega} F \cdot d\mathbf{l}$$

Divergence Theorem (Space) Let $\Omega \subseteq \mathbb{R}^3$ be a regular domain, $n : \partial\Omega \rightarrow \mathbb{R}^3$ a continuous outward unit normal vector field, and $F \in C^1(\overline{\Omega}, \mathbb{R}^3)$. Then

Divergence Theorem (Plane) Let $\Omega \subseteq \mathbb{R}^2$ be a regular domain with a positively oriented boundary $\partial\Omega$, and $F \in C^1(\overline{\Omega}, \mathbb{R}^2)$. Then

$$\iint_{\Omega} \operatorname{div} F(x, y) dx dy = \int_{\partial\Omega} F \cdot n dl = \int_a^b \langle F(\gamma(t)), (\gamma'_2(t), -\gamma'_1(t)) \rangle dt.$$

Stokes' Theorem Let $\Omega \subseteq \mathbb{R}^3$ be an open set, $\Sigma \subseteq \Omega$ a piecewise smooth orientable surface, and $F \in C^1(\Omega, \mathbb{R}^3)$, then

$$\iint_{\Sigma} \operatorname{curl} F \cdot n ds = \int_{\partial\Sigma} F \cdot d\mathbf{l} = \iint_A \left\langle \operatorname{curl} F(\sigma(u, v)), \frac{\partial \sigma}{\partial u} \wedge \frac{\partial \sigma}{\partial v} \right\rangle du dv$$

Polar (2D)

$$\begin{aligned} |\det J| &= r \\ \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} r \cos \theta \\ r \sin \theta \end{bmatrix}; \\ r &\geq 0, \theta \in [0, 2\pi]. \end{aligned}$$

Cylindrical (3D)

$$\begin{aligned} |\det J| &= r \\ \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} r \cos \theta \\ r \sin \theta \\ z \end{bmatrix}; \end{aligned}$$

Spherical (3D)

$$\begin{aligned} |\det J| &= \rho^2 \sin \phi \\ \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} \rho \sin \phi \cos \theta \\ \rho \sin \phi \sin \theta \\ \rho \cos \phi \end{bmatrix}; \end{aligned}$$

Distribution Theory

Let $\mathcal{D}'$ be the set of distributions over $\mathbb{R}$, the set of linear continuous functionals over $\mathcal{D}$, $\mathcal{D}' = \{T : \mathcal{D} \rightarrow \mathbb{R} \mid T \text{ is linear and continuous}\}$. For a distribution $T \in \mathcal{D}'$ and a test function $\varphi \in \mathcal{D}$, the pairing is defined by $\langle T, \varphi \rangle = \int_{\Omega} f(x) \varphi(x) dx$.

A distribution $T \in \mathcal{D}'$ satisfies:

Support

$$\operatorname{supp}(T) = \overline{\{x \in \mathbb{R} \mid T(\varphi) \neq 0\}}.$$

Derivative

$$\varphi \in \mathcal{D}, T'(\varphi) = -T(\varphi').$$

Boundedness For every $\psi \in \mathcal{D}$, $|T(\psi)|$ is finite.

Linearity For all scalars $\alpha, \beta \in \mathbb{R}$ and test functions $\psi, \varphi \in \mathcal{D}$, $T(\alpha\psi + \beta\varphi) = \alpha T(\psi) + \beta T(\varphi)$.

Continuity

$\forall [a, b] \subseteq \mathbb{R}$, there exist constants $C > 0$ and $k \in \mathbb{N}_0$, such that $\forall \varphi \in \mathcal{D}$, $\operatorname{supp}(\varphi) \subseteq [a, b] \implies |T(\varphi)| \leq C \sum_{0 \leq i \leq k} \max_{x \in \mathbb{R}} |\partial^i \varphi(x)|$.

Higher-Order Derivatives $\forall n \in \mathbb{N}$, $T^{(n)}(\varphi) = (-1)^n T(\varphi^{(n)})$.

Dirac Delta

$$\begin{aligned} \delta(\varphi) &= \varphi(0) \text{ for all } \varphi \in \mathcal{D}. \\ \delta'(\varphi) &= -\varphi'(0) \text{ for all } \varphi \in \mathcal{D}. \end{aligned}$$

Dirac Comb

$$\begin{aligned} \Delta_T(x) &= \sum_{n \in \mathbb{Z}} \delta(x - nT), \\ T > 0 \text{ is the period.} \end{aligned}$$

Scaled Dirac Delta

$$\begin{aligned} a \in \mathbb{R}, (a\delta)(\varphi) &= a \cdot \varphi(0). \\ (a\delta)'(\varphi) &= -a \cdot \varphi'(0). \end{aligned}$$

Heaviside Step $H'(x) = \delta(x)$.

$$H(x) = \begin{cases} 0 & \text{if } x < 0, \\ 1 & \text{if } x \geq 0. \end{cases}$$

Regular Distribution

For any (locally) integrable function f on $\mathbb{R}$:

- $\langle f, \varphi \rangle := \int_{\mathbb{R}} f(x) \varphi(x) dx$.
- 1. *Finiteness*: $|\langle f, \varphi \rangle| \leq \int_{\mathbb{R}} |f(x)| |\varphi(x)| dx \leq C \max_{x \in \mathbb{R}} |\varphi(x)|$, $C := \int_{\mathbb{R}} |f(x)| dx$.
- 2. *Linearity*: For scalars α, β and $\varphi, \psi \in \mathcal{D}$: $\langle f, \alpha\varphi + \beta\psi \rangle = \alpha \langle f, \varphi \rangle + \beta \langle f, \psi \rangle$.
- 3. *Continuity*: $|\langle f, \varphi \rangle| \leq C \max_{x \in \mathbb{R}} |\varphi(x)|$.

Let $\Omega \subseteq \mathbb{R}^n$.

Convex if $\forall x, y \in \Omega$, the line segment from x to y is within Ω .

Star-shaped if $\exists z \in \Omega$ such that $\forall x \in \Omega$, the line segment from z to x is within Ω . $\operatorname{rot}(\operatorname{rot} \vec{F}) = -\Delta \vec{F} + \operatorname{grad}(\operatorname{div} \vec{F})$

$$\begin{aligned} \operatorname{div}(f \operatorname{grad} g) &= f \Delta g + \nabla f \cdot \nabla g \\ \nabla \cdot (f \vec{A}) &= (\nabla f) \cdot \vec{A} + f(\nabla \cdot \vec{A}) \\ \Delta(fg) &= f \Delta g + 2 \nabla f \cdot \nabla g + g \Delta f \end{aligned}$$

Piecewise Continuity & Differentiability
 $f : [a, b] \rightarrow \mathbb{R}$ is *piecewise continuous* if there is a partition

$$a = a_0 < a_1 < \cdots < a_n = b$$

such that $\lim_{x \rightarrow a_i^-} f(x)$ and $\lim_{x \rightarrow a_i^+} f(x)$ exist (finite).

Similarly, f is *piecewise C^1* if it is continuously differentiable on each open subinterval and the one-sided derivatives at boundaries exist.

Euler’s Formulas

$$e^{x+iy} = e^x (\cos y + i \sin y), \quad \sin x = \frac{e^{ix} - e^{-ix}}{2i}, \quad \cos x = \frac{e^{ix} + e^{-ix}}{2}.$$

Orthogonality (Sine/Cosine Products)

For $n, m \in \mathbb{N}_{\geq 1}$ and period $T > 0$:

$$\frac{2}{T} \int_0^T \cos\left(\frac{2\pi n}{T}x\right) \cos\left(\frac{2\pi m}{T}x\right) dx = \begin{cases} 1 & n = m, \\ 0 & n \neq m \end{cases}$$

(Same for $\sin \sin$, and $\cos \sin$ integrates to 0.)

Integration Over One Period

If f is T -periodic and piecewise continuous, then for any $a \in \mathbb{R}$:

$$\int_a^{a+T} f(x) dx = \int_0^T f(x) dx.$$

Dirichlet’s Theorem (Pointwise Convergence)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be T -periodic and piecewise C^1 . Then, for all $x \in \mathbb{R}$,

$$Ff(x) = \lim_{t \rightarrow 0} \frac{f(x-t) + f(x+t)}{2}.$$

Real Fourier Series

For $f : \mathbb{R} \rightarrow \mathbb{R}$, T -periodic, piecewise C^1 , the real Fourier series is

$$Ff(x) = \frac{a_0}{2} + \sum_{n=1}^\infty \left[a_n \cos\left(\frac{2\pi n}{T}x\right) + b_n \sin\left(\frac{2\pi n}{T}x\right) \right].$$

Fourier Coefficients:

$$a_n = \frac{2}{T} \int_0^T f(x) \cos\left(\frac{2\pi n}{T}x\right) dx, \quad b_n = \frac{2}{T} \int_0^T f(x) \sin\left(\frac{2\pi n}{T}x\right) dx, \\ a_0 = \frac{2}{T} \int_0^T f(x) dx.$$

Parity: If f is even, $b_n = 0$; if f is odd, $a_n = 0$.

Term-by-Term Differentiation

If f is T -periodic, continuous, and piecewise C^1 , then

$$\frac{d}{dx} [Ff(x)] = \sum_{n=1}^\infty \frac{2\pi n}{T} [-a_n \sin(\frac{2\pi n}{T}x) + b_n \cos(\frac{2\pi n}{T}x)] \\ = \lim_{t \rightarrow 0} \frac{f'(x-t) + f'(x+t)}{2}.$$

Term-by-Term Integration

If f is T -periodic, continuous, and piecewise C^1 , then

$$\int Ff(x) dx = \sum_{n=1}^\infty \frac{T}{2n\pi} \left[a_n \sin\left(\frac{2\pi n}{T}x\right) - b_n \cos\left(\frac{2\pi n}{T}x\right) \right] + C \\ = \lim_{h \rightarrow 0} \frac{1}{2h} \int_{x-h}^{x+h} Ff(t) dt,$$

where C is the constant of integration.

Poisson on $[a, b]$

$$\begin{cases} -u''(x) = f(x), \\ u(a) = g_a, \quad u(b) = g_b, \end{cases} \quad L = b - a$$

$$u^g(x) = \frac{g_b - g_a}{b - a} x + \frac{b g_a - a g_b}{b - a}.$$

$$f(x) = \sum_{n=1}^\infty b_n \sin\left(\frac{n\pi x}{L}\right), \quad b_n = \frac{2}{L} \int_0^L f(t) \sin\left(\frac{n\pi t}{L}\right) dt,$$

$$u^f(x) = \sum_{n=1}^\infty b_n \frac{L^2}{\pi^2 n^2} \sin\left(\frac{n\pi x}{L}\right).$$

$$u(x) = u^g(x) + u^f(x) \quad (\text{superposition principle}).$$

Poisson with mass term on $\mathbb{R}$

$$-u''(x) + k^2 u(x) = f(x), \quad \widehat{u}(\alpha) = \frac{\widehat{f}(\alpha)}{\alpha^2 + k^2}.$$

$$g(x) = \sqrt{\frac{\pi}{2k}} e^{-k|x|}, \quad \widehat{g}(\alpha) = \frac{1}{\alpha^2 + k^2},$$

$$u(x) = (g * f)(x) = \frac{1}{2k} \int_{-\infty}^\infty f(y) e^{-k|x-y|} dy.$$

Complex Fourier Coefficient

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be T -periodic and piecewise continuous. The complex Fourier coefficients are:

$$c_n = \frac{1}{T} \int_0^T f(x) e^{-i\frac{2\pi}{T}nx} dx, \quad Ff(x) = \sum_{n=-\infty}^\infty c_n e^{i\frac{2\pi n}{T}x}.$$

For $\phi : \mathbb{R} \rightarrow \mathbb{C}$,

$$\int_a^b \phi(x) dx = \int_a^b \operatorname{Re}(\phi(x)) dx + i \int_a^b \operatorname{Im}(\phi(x)) dx.$$

Relation to (a_n, b_n)

$$c_n = \frac{1}{2}(a_n - i b_n), \quad c_{-n} = \frac{1}{2}(a_n + i b_n), \quad c_0 = \frac{a_0}{2}. \\ a_n = c_n + c_{-n} \quad a_0 = 2c_0 \quad b_n = i(c_n - c_{-n})$$

Fourier Series on $[0, L]$

For $f : [0, L] \rightarrow \mathbb{R}$ (piecewise C^1):

$$F_c f(x) = \frac{\tilde{a}_0}{2} + \sum_{n=1}^\infty \tilde{a}_n \cos\left(\frac{\pi n}{L}x\right), \quad \tilde{a}_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{\pi n}{L}x\right) dx.$$

$$F_s f(x) = \sum_{n=1}^\infty \tilde{b}_n \sin\left(\frac{\pi n}{L}x\right), \quad \tilde{b}_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{\pi n}{L}x\right) dx.$$

Parseval’s Identity (Periodic Case)

If f is T -periodic (piecewise C^1),

$$\frac{2}{T} \int_0^T f^2(x) dx = \frac{a_0^2}{2} + \sum_{n=1}^\infty (a_n^2 + b_n^2) = 2 \sum_{n=-\infty}^\infty |c_n|^2.$$

Plancherel Theorem

Let $f \in L^2(\mathbb{R})$. Then its Fourier transform $\hat{f}$ is also in $L^2(\mathbb{R})$, and:

$$\int_{-\infty}^\infty |f(x)|^2 dx = \int_{-\infty}^\infty |\hat{f}(\xi)|^2 d\xi.$$

The Fourier Transform

If $f : \mathbb{R} \rightarrow \mathbb{R}$ with $\int_{-\infty}^\infty |f(x)| dx < \infty$, its (unitary) Fourier transform is

$$\hat{f}(\alpha) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty f(x) e^{-i\alpha x} dx,$$

Inverse Transform

If $\varphi(\alpha)$ is similarly integrable,

$$\mathcal{F}^{-1}(\varphi)(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty \varphi(\alpha) e^{i\alpha x} d\alpha.$$

Convolution Product

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ such that $\int_{-\infty}^{+\infty} |f(x)| dx < +\infty$, $\int_{-\infty}^{+\infty} |g(x)| dx < +\infty$.

$$(f * g)(x) = \int_{-\infty}^{+\infty} f(x-t) g(t) dt = \int_{-\infty}^{+\infty} f(t) g(x-t) dt$$

Scaling

$$\mathcal{F}\{f(ax)\} = \frac{1}{|a|} \hat{f}\left(\frac{\alpha}{a}\right)$$

Shifting

$$\mathcal{F}\{f(x - x_0)\} = e^{-i\alpha x_0} \hat{f}(\alpha)$$

Modulation

$$\mathcal{F}\{e^{i\omega_0 x} f(x)\} = \hat{f}(\alpha - \omega_0)$$

Convolution

$$\mathcal{F}[f * g](\alpha) = \sqrt{2\pi} \hat{f}(\alpha) \hat{g}(\alpha)$$

Differentiation

$$\mathcal{F}\left\{\frac{d^n}{dx^n} f(x)\right\} = (i\alpha)^n \hat{f}(\alpha)$$

Integration

$$\mathcal{F}\left\{\int_{-\infty}^x f(\xi) d\xi\right\} = \frac{1}{i\alpha} \hat{f}(\alpha), \quad \alpha \neq 0$$

Differentiation of Transform

$$\mathcal{F}((-ix)^n f(x))(\alpha) = \frac{\partial^n}{\partial \alpha^n} \hat{f}(\alpha)$$

Multiplication

$$\mathcal{F}\{f(x) \cdot g(x)\} = \sqrt{2\pi} (\mathcal{F}\{f(x)\} * \mathcal{F}\{g(x)\})$$

Important Trigonometric Identities

$$\sin(2x) = 2 \sin x \cos x,$$

$$\cos(2x) = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x = \cos^2 x - \sin^2 x,$$

$$\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b,$$

$$\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b,$$

$$\cos a \cos b = \frac{1}{2} [\cos(a - b) + \cos(a + b)],$$

$$\sin a \sin b = \frac{1}{2} [\cos(a - b) - \cos(a + b)],$$

$$\sin a \cos b = \frac{1}{2} [\sin(a + b) + \sin(a - b)],$$

$$\cos a \sin b = \frac{1}{2} [\sin(a + b) - \sin(a - b)].$$

$$\cos(n\pi) = (-1)^n$$